Lecture–2: Introduction to Pandas

Topics Covered:

1. Introduction to Pandas
2. Important List of Operations in Pandas
3. Installing and Importing Pandas
4. Creating Series Data Structure
5. Update / Delete / Insert Operations on Pandas Series

(According to your Placement Oriented DA syllabus – Lecture 2)

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1. What is Pandas?

Definition

Pandas is an open-source Python library used for:

- Reading datasets
- Cleaning data
- Transforming data
- Filtering data
- Analyzing data
- Statistical operations

Pandas mainly works with:

1. Series → 1D data
2. DataFrame → 2D tabular data

Why Pandas is Needed?

Suppose Amazon has:

Customer	City	Purchase
Ram	Delhi	1000
Shyam	Mumbai	2000

Customer	City	Purchase
Rita	Pune	3000

Questions:

- Average purchase?
- Highest purchase?
- Customers from Delhi?
- Missing values?

Pandas solves such problems easily.

Why Learn Pandas for Placements?

Companies ask:

- Amazon
- TCS
- Accenture
- Deloitte
- Infosys
- Flipkart

Typical interview questions:

"How to handle missing values?"

"How to read CSV?"

"Difference between loc and iloc?"

"GroupBy usage?"

Features of Pandas

Pandas supports:

✓ Fast operations

✓ Handle missing data

✓ Read multiple file formats

CSV

Excel

JSON

SQL

✓ Statistical analysis

✓ Data cleaning

✓ Grouping

✓ Merging datasets

Pandas Data Structures

There are two major structures:

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1. Series (1D)

Looks like:

Index	Value
0	10
1	20
2	30

Similar to:

[10,20,30]

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2. DataFrame (2D)

Looks like:

Name	Age
Ram	22
Rita	25

Similar to Excel table.

Real-world Analogy

Series:

One subject marks

Math Marks:
80
90
70

DataFrame:

Complete report card

Name	Math	Science
Ram	80	90

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2. Important Operations in Pandas

Interview and placement important operations:

Data Reading

Read CSV:

pd.read_csv()

Read Excel:

pd.read_excel()

Read JSON:

pd.read_json()

Data Viewing

View top rows:

df.head()

View last rows:

df.tail()

Shape:

df.shape

Columns:

df.columns

Statistical Operations

Mean:

df.mean()

Maximum:

df.max()

Minimum:

df.min()

Count:

df.count()

Missing Value Handling

Check null:

df.isnull()

Total null:

df.isnull().sum()

Remove null:

df.dropna()

Fill null:

df.fillna()

Filtering

Condition:

df[df["Age"]>25]

Sorting

Ascending:

df.sort_values()

Descending:

df.sort_values(ascending=False)

Grouping

df.groupby()

Example:

Average salary department-wise.

Merging

Join two tables:

pd.merge()

Most Important Pandas Functions for Placements

Remember:

head()
tail()
shape
info()
describe()
loc[]
iloc[]
groupby()
merge()
fillna()
dropna()
sort_values()

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3. Installing Pandas

Check Python Installed

Open terminal:

python --version

Output:

Python 3.x.x

Install Pandas

Using pip:

pip install pandas

Install specific version:

pip install pandas==2.0

Update pandas:

pip install --upgrade pandas

Check installation:

pip show pandas

Import Pandas

Standard syntax:

import pandas as pd

Why "pd"?

Alias

Instead of:

pandas.Series()

Write:

pd.Series()

Test import:

import pandas as pd

print("Pandas imported")

Output:

Pandas imported

Check version:

import pandas as pd

print(pd.__version__)

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4. Creating Series Data Structure

What is Series?

Series = One-dimensional labeled array

Contains:

Value + Index

Example:

Index	Value
0	10
1	20
2	30

Create Series from List

import pandas as pd

```
x = [10,20,30]

s = pd.Series(x)

print(s)
```

Output:

```
0    10
1    20
2    30
dtype:int64
```

Explanation:

Index automatically assigned.

Create Series with Custom Index

```
import pandas as pd

x=[10,20,30]

s=pd.Series(x,index=["A","B","C"])

print(s)
```

Output:

```
A 10
B 20
C 30
```

Create Series from String List

```
name=["Ram","Shyam","Rita"]

s=pd.Series(name)

print(s)
```

Output:

0 Ram
1 Shyam
2 Rita

Create Series from Dictionary

```
student={
"Math":90,
"Science":80
}

s=pd.Series(student)

print(s)
```

Output:

```
Math       90
Science    80
```

Create Series with Single Value

```
s=pd.Series(100,index=[1,2,3])

print(s)
```

Output:

```
1 100
2 100
3 100
```

Access Series Values

Suppose:

```
s=pd.Series([10,20,30])
```

Access:

```
print(s[0])
```

Output:

```
10
```

Slice Series

print(s[0:2])

Output:

10
20

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5. Update / Insert / Delete Operations on Series

Assume:

import pandas as pd

s=pd.Series([10,20,30])

print(s)

Output:

0 10
1 20
2 30

Update Value

Change 20 → 200

s[1]=200

print(s)

Output:

0 10
1 200
2 30

Real-world:

Student marks updated.

Insert New Value

Append:

s[3]=40

print(s)

Output:

0 10
1 20
2 30
3 40

Insert with label:

s["Math"]=90

Output:

Math 90

Delete Value

Delete index:

s=s.drop(1)

print(s)

Output:

0 10
2 30

Explanation:

Index 1 removed.

Delete Multiple Values

s=s.drop([0,2])

Delete Entire Series

del s

Example: Student Marks

Create:

```
marks=pd.Series(
[80,90,70],
index=["Ram","Rita","Shyam"]
)

print(marks)
```

Output:

```
Ram 80
Rita 90
Shyam 70
```

Update:

```
marks["Ram"]=100
```

Insert:

```
marks["Aman"]=95
```

Delete:

```
marks=marks.drop("Shyam")
```

Final:

```
Ram   100
Rita   90
Aman   95
```

Common Interview Questions

Q1 What is Pandas?

Python library for data manipulation and analysis.

Q2 Difference between Series and DataFrame?

Series:

1D

DataFrame:

2D

Q3 Why import pandas as pd?

Alias for shorter code.

Q4 How to create Series?

pd.Series()

Q5 How to remove values from Series?

drop()

Lecture–2 Summary

Students should know:

✓ What is Pandas
✓ Why Pandas is important

✓ Series vs DataFrame

✓ Installing pandas

✓ Importing pandas

✓ Creating Series

✓ Updating Series

✓ Inserting values

✓ Deleting values

✓ Basic Pandas operations

Homework Practice

Question 1

Create Series:

10
20
30
40

Question 2

Create Series using names:

Ram
Shyam
Rita

Question 3

Update:

20 → 200

Question 4

Insert:

50

Question 5

Delete:

30